

EDA USING POWER BI

REPORT

WEEK – 2

**DATA MODELING & DASHBOARD
DEVELOPMENT – SALES DATASET**

SUBMITTED BY

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1. Dataset:

Retail Sales Dataset (Preprocessed from Week 1)

2. Problem Statement:

The organization needs an interactive dashboard to track sales revenue, profit, product performance, customer segments, and yearly sales trends.

3. Objective:

To develop a structured data model and interactive Power BI dashboard for analyzing retail sales performance.

4. Tools Used:

Power BI Desktop

Retail Sales Dataset

5.Steps Performed:

Step 1: Data Cleaning & Dimensional Modeling

Created dimension tables from the main dataset:

Dim_Product – Product ID, Product Name, Category, Sub-Category

Dim_Customer – Customer ID, Customer Name,
Segment

Dim_Location – Region, State, City

Dim_Date – Order Date, Year, Month, Month Name

Main Fact Table: Retail Sales

25 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 9:34 AM

Queries [3] = Table.Distinct(#"Removed Columns", {"Product Name"})

	Product ID	Category	Sub-Category	Product Name
	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%
1	OFF-PA-10000174	Office Supplies	Paper	Message Book, Wirebound, Four 5 1/2" X 4" Forms/Pg., 200 Dupl. Sets...
2	OFF-BI-10004094	Office Supplies	Binders	GBC Standard Plastic Binding Systems Combs
3	OFF-LA-10003223	Office Supplies	Labels	Avery 508
4	OFF-ST-10002743	Office Supplies	Storage	SAFCO Boltless Steel Shelving
5	OFF-AR-10003478	Office Supplies	Art	Avery Hi-Liter EverBold Pen Style Fluorescent Highlighters, 4/Pack
6	FUR-CH-10004063	Furniture	Chairs	Global Deluxe High-Back Manager's Chair
7	OFF-AR-10001662	Office Supplies	Art	Rogers Handheld Barrel Pencil Sharpener
8	OFF-AR-10002399	Office Supplies	Art	Dixon Prang Watercolor Pencils, 10-Color Set with Brush
9	OFF-BI-10004632	Office Supplies	Binders	Ibico Hi-Tech Manual Binding System
10	OFF-FA-10001883	Office Supplies	Fasteners	Alliance Super-Size Bands, Assorted Sizes
11	OFF-PA-10000955	Office Supplies	Paper	Southworth 25% Cotton Granite Paper & Envelopes
12	OFF-PA-10002005	Office Supplies	Paper	Xerox 225
13	TEC-PH-10004539	Technology	Phones	Wireless Extenders zBoost YX545 SOHO Signal Booster
14	TEC-PH-10004977	Technology	Phones	GE 30524EE4
15	FUR-FU-10004864	Furniture	Furnishings	Howard Miller 14-1/2" Diameter Chrome Round Wall Clock
16	OFF-BI-10003708	Office Supplies	Binders	Acco Four Pocket Poly Ring Binder with Label Holder, Smoke, 1"
17	OFF-AR-10004078	Office Supplies	Art	Newell 312
18	TEC-AC-10001266	Technology	Accessories	Memorex Micro Travel Drive 8 GB
19	FUR-FU-10000965	Furniture	Furnishings	Howard Miller 11-1/2" Diameter Ridgewood Wall Clock
20	OFF-LA-10004272	Office Supplies	Labels	Avery 482
21	FUR-FU-10000010	Furniture	Furnishings	DAX Value U-Channel Document Frames, Easel Back
22	FUR-BO-10003034	Furniture	Bookcases	O'Sullivan Elevations Bookcase, Cherry Finish
23	FUR-CH-10000422	Furniture	Chairs	Global Highback Leather Tilter in Burgundy
24	OFF-AR-10003514	Office Supplies	Art	Avery Washable Dry Erase

Step 2: Created Measures Table (_Measures)

Created important DAX measures:

Total Revenue = SUM(Sales[Revenue])

Total Profit = SUM(Sales[Profit])

Total Quantity = SUM(Sales[Quantity])

Profit Percentage = DIVIDE([Total Profit], [Total Revenue])

Create Table

	Column1	+
1		
+		

Name:

Measures table has been created.

The screenshot displays the Microsoft Power BI Desktop interface. The 'Table tools' ribbon is active, showing options for 'Date Table', 'Manage relationships', 'New visual calculation', 'New measure', 'Quick measure', 'New measure column', 'New table', and 'Mark as date table'. The 'Date Table' is selected, and its DAX formula is shown: `1 Date Table = CALENDAR(MIN('SHOP_2015-05-21_2025-03-16'[date]),MAX('SHOP_2015-05-21_2025-03-16'[date]))`. The 'Visualizations' pane on the right shows the 'Date Table' selected, with a list of fields including Date, Date Hierarchy, Date, Month, Quarter, Year, Day, Month Number, Quarter, and Year. The 'Measures (2)' section is also visible, listing fields like Close Price, Close Price PY, Close Price YoY, Column1, Daily Return %, High Price, Low Price, Open Price, Previous Close, Trading Volume, and Trading Volume...

Step 3: Data Model Creation

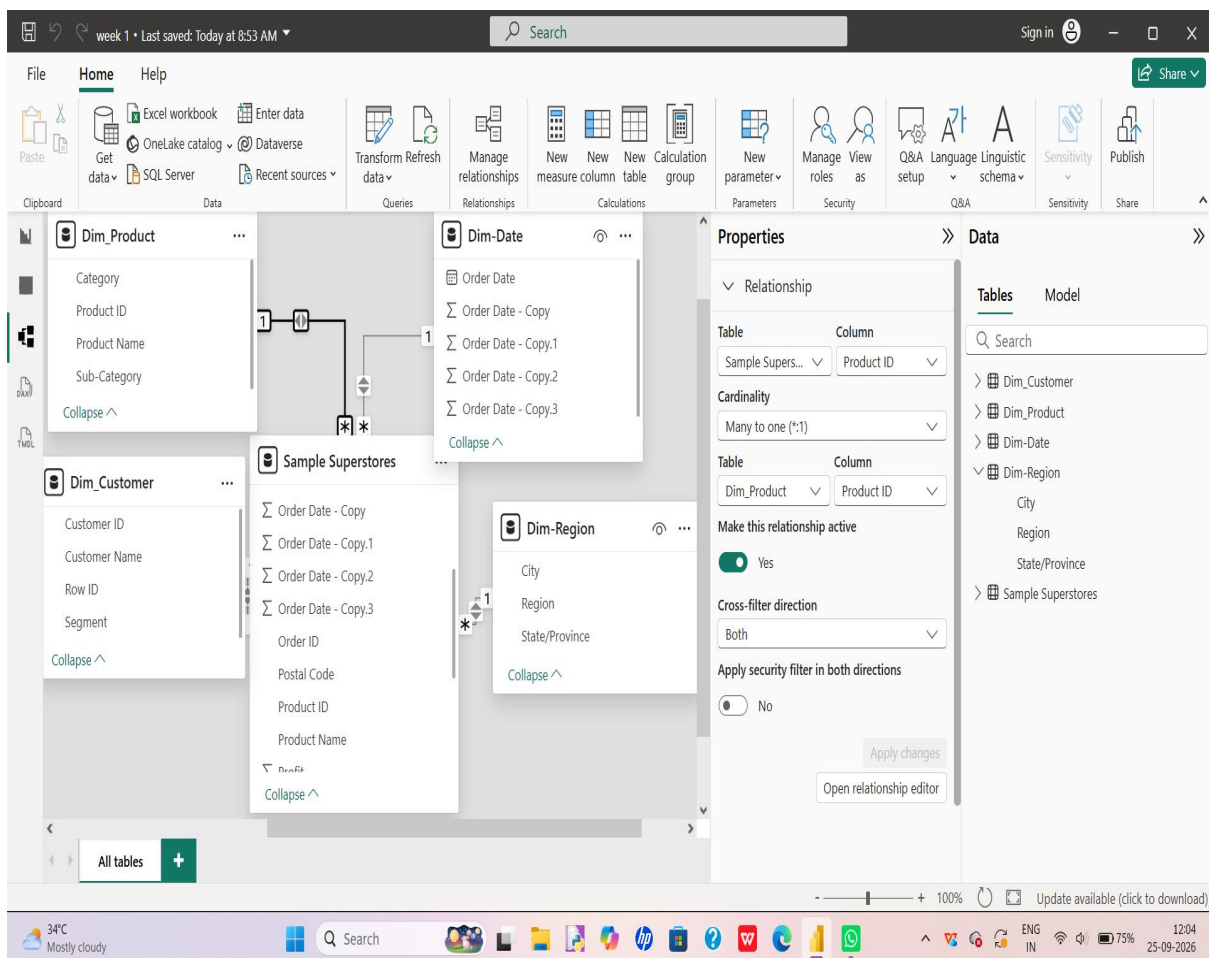
Established one-to-many relationships between the dimension tables and the main sales table.

Dim_Product → Retail Sales

Dim_Customer → Retail Sales

Dim_Date → Retail Sales

Dim_Location → Retail Sales



Step 4: Dashboard Creation

KPI CARDS:

Total Revenue

Total Profit

Total Quantity

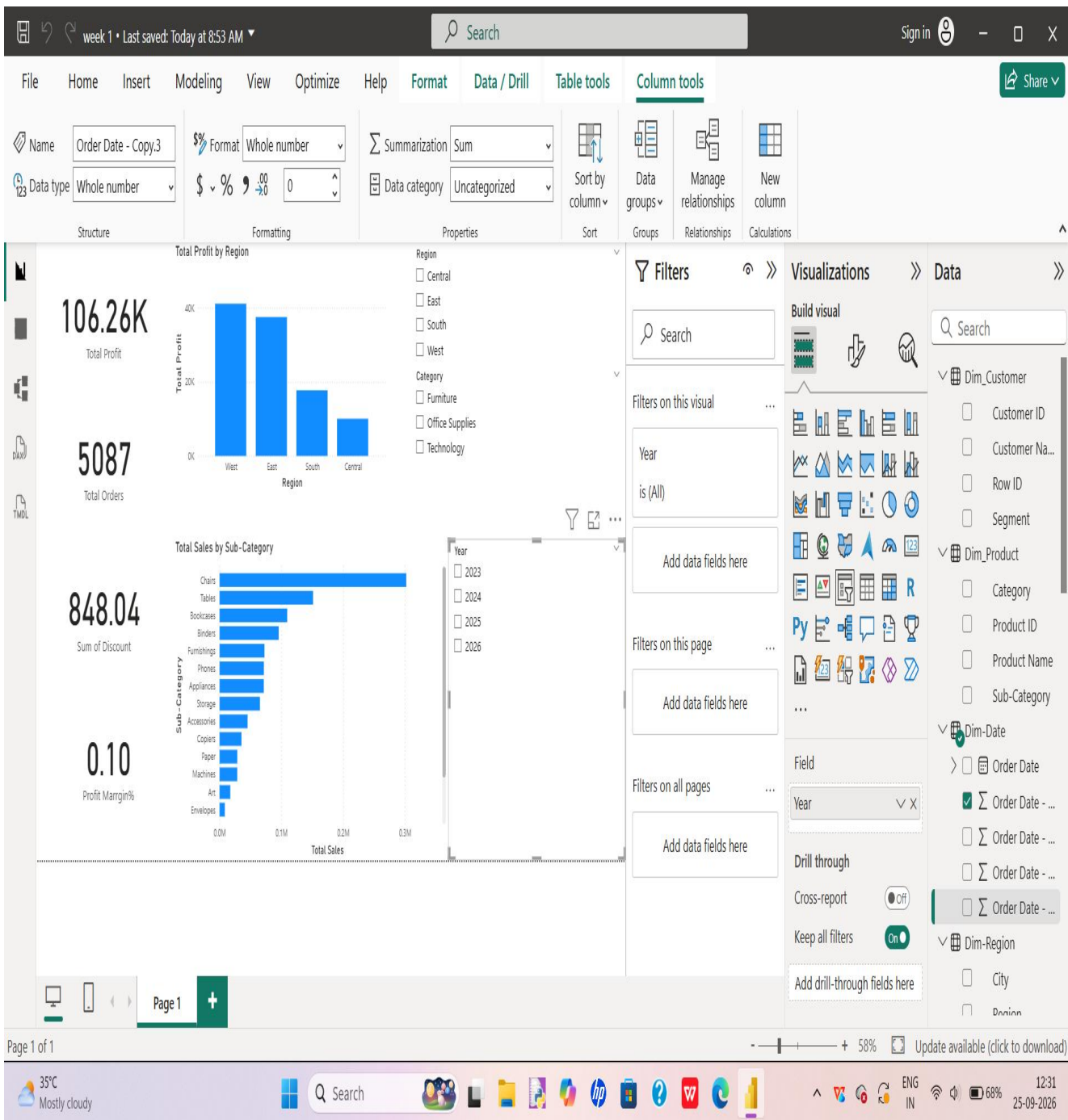
Profit Percentage

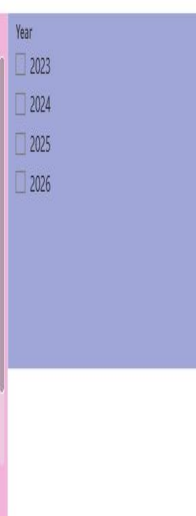
Visuals:

- Column Chart: Total Sales by Category
(X-axis: Category, Y-axis: Total Sales)
- Bar Chart: Total Sales by Sub-Category
(Y-axis: Sub-Category, X-axis: Total Sales)
- Line Chart: Total Sales by Year
(X-axis: Year, Y-axis: Total Sales)
- Slicer: Category
(Used to filter the dashboard based on product categories)
- KPI Cards: Total Sales, Total Orders, Total Profit, and Profit Margin

Step 5:

Interactive Sales Dashboard Creation





Filters

Search

Filters on this page ...

Add data fields here

Filters on all pages ...

Add data fields here

Visualizations

Build visual

...

Values

Drill through

Cross-report
☐ Off

Keep all filters
☒ On

Data

Search

- > Dim_Customer
- > Dim_Product
- > Dim-Date
- > Dim-Region
- > Measures (2)
- > Sample Superstores

week 1 • Last saved: Today at 8:53 AM

Search

Sign in

FileHomeInsertModelingViewOptimizeHelpFormatData / Drill

CutCopyFormat painterClipboard

Get dataExcelOneLakeSQLEnter dataDataverseRecent sourcesData

Transform Refresh dataQueries

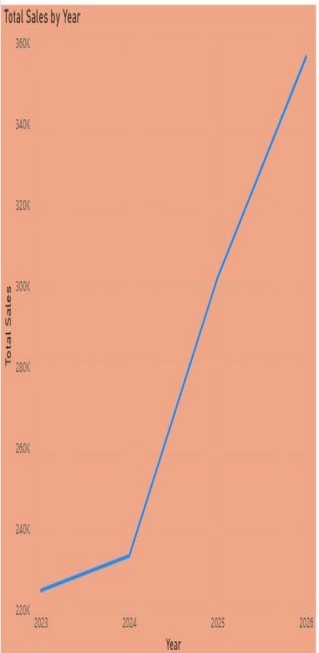
New visualText boxMore visualsInsert

New visual calculationNew Quick measure measureCalculations

SensitivitySensitivityShare

Prep data for Copilot AICopilot

Total Sales by Year



Sub-Category



Filters

Search

Filters on this visual

Category is (All)

Region is (All)

Sub-Category is (All)

Total Sales is (All)

Add data fields here

Filters on this page

Visualizations

Format visual

Search

Visual General

Properties

Title

Effects

Background

Color

Transparency

Visual border

Data

Search

☐ Order Date - ...

☐ Order Date - ...

☐ Order Date - ...

☐ Order Date - ...

☒ Dim-Region

☐ City

☒ Region

☐ State/Province

☒ Measures (2)

☐ Column1

☐ Profit Margin%

☐ Total Orders

☐ Total Profit

☒ Total Sales

☒ Sample Superstores

☐ Category

☐ City

☐ Country/Region

☐ Customer ID

Page 2 of 2

Page 1Page 2

58%

Update available (click to download)

35°C Mostly cloudy

Search

hp

ENG IN

65%

12:40 25-09-2026

6. Conclusion:

The Power BI dashboard provides an interactive view of sales performance. The data model helps organize information efficiently, while DAX measures provide important business calculations. The dashboard makes it easier to compare products, regions, customers, and sales trends.